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Systems for skin patch gravity resistance

Abstract

A system for a physiological characteristic sensor deployed with a sensor inserter includes an adhesive skin patch coupled to the physiological characteristic sensor. The adhesive patch is to couple the physiological characteristic sensor to an anatomy. The system also includes a gravity resistance system coupled to the adhesive patch and to be coupled to the sensor inserter. The gravity resistance system maintains the adhesive patch substantially perpendicular to a longitudinal axis of the sensor inserter prior to deployment of the physiological characteristic sensor and is removable from the adhesive patch by the sensor inserter upon deployment of the physiological characteristic sensor.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 16/719,892, filed Dec. 18, 2019, of which is incorporated herein by reference in their entirety.

FIELD

(1) Embodiments of the subject matter described herein relate generally to medical devices, such as a skin patch for a physiological characteristic sensor assembly. More particularly, embodiments of the subject matter relate to systems that improve gravity resistance of a skin patch during storage to ensure that the skin patch remains ready for coupling to a user after a period of time.

BACKGROUND

(2) Sensors may be employed in the treatment of or monitoring of various medical conditions. In one example, thin film electrochemical sensors are used to test analyte levels in patients or users. More specifically, thin film sensors have been designed for use in obtaining an indication of blood glucose (BG) levels and monitoring BG levels in a diabetic user, with the distal segment portion of the sensor positioned subcutaneously in direct contact with extracellular fluid. Such readings can be especially useful in adjusting a treatment regimen which typically includes regular administration of insulin to the user.

(3) A glucose sensor of the type described above may be packaged and sold as a product, such as a continuous glucose monitor, which is adhered to the patient during use via an adhesive skin patch. In certain instances, the continuous glucose monitor may be packaged with a sensor introducer tool, which enables the implantation of the glucose sensor subcutaneously/transcutaneously. The sensor introducer tool contains a needle that is used to puncture the skin of a user at the same time as the sensor is introduced. The needle is then withdrawn, leaving the sensor in the skin of the user.

(4) In instances where the continuous glucose sensor is packaged with the sensor introducer tool, the continuous glucose sensor may be positioned within the sensor introducer tool such that the skin patch is subjected to the effects of gravity. Gravity, when acting on the skin patch, may cause

the skin patch to droop or sag within the sensor introducer tool. When the skin patch droops or sags within the sensor introducer tool, the skin patch may fold upon itself, and thus, may not adhere well to the user.

(5) Accordingly, it is desirable to provide systems for improving gravity resistance of a skin patch, such as a skin patch coupled to a physiological characteristic sensor, for example, a glucose sensor or continuous glucose monitor, which inhibits the skin patch from drooping or sagging to ensure that the skin patch remains ready for coupling to a user after a period of time. Furthermore, other desirable features and characteristics will become apparent from the subsequent detailed description and the appended claims, taken in conjunction with the accompanying drawings and the foregoing technical field and background.

BRIEF SUMMARY

(6) The techniques of this disclosure generally relate to systems that improve gravity resistance of an adhesive skin patch, such as an adhesive skin patch coupled to a medical device, such as a glucose sensor or continuous glucose monitor.

(7) Provided according to various embodiments is a system for a physiological characteristic sensor deployed with a sensor inserter. The system includes an adhesive patch coupled to the physiological characteristic sensor. The adhesive patch is to couple the physiological characteristic sensor to an anatomy. The system also includes a gravity resistance system coupled to the adhesive patch and to be coupled to the sensor inserter. The gravity resistance system maintains the adhesive patch substantially perpendicular to a longitudinal axis of the sensor inserter prior to deployment of the physiological characteristic sensor and is removable from the adhesive patch by the sensor inserter upon deployment of the physiological characteristic sensor.

(8) Also provided is a system for a physiological characteristic sensor deployed with a sensor inserter. The system includes an adhesive patch coupled to the physiological characteristic sensor. The adhesive patch is to couple the physiological characteristic sensor to an anatomy. The system includes a gravity resistance system coupled to the adhesive patch and to the sensor inserter. The gravity resistance system includes at least one adhesive layer coupled between the adhesive patch and the sensor inserter. The at least one adhesive layer is coupled to a surface of the adhesive layer so as to be positioned about at least a portion of a perimeter of the adhesive patch. The gravity resistance system maintains the adhesive patch substantially perpendicular to a longitudinal axis of the sensor inserter prior to deployment of the physiological characteristic sensor and is removable from the adhesive patch by the sensor inserter upon deployment of the physiological characteristic sensor.

(9) Further provided is a system for a physiological characteristic sensor deployed with a sensor inserter. The system includes an adhesive patch coupled to the physiological characteristic sensor. The adhesive patch is to couple the physiological characteristic sensor to an anatomy. The system includes a gravity resistance system coupled to the adhesive patch and to the sensor inserter. The gravity resistance system includes at least one adhesive layer coupled between the adhesive patch and the sensor inserter. The at least one adhesive layer is coupled to a surface of the adhesive layer so as to be positioned about a perimeter of the adhesive patch. The at least one adhesive layer comprises a first tack adhesive on a first side and a second tack adhesive on an opposite side, and the second tack adhesive is less tacky than the first tack adhesive. The gravity resistance system maintains the adhesive patch substantially perpendicular to a longitudinal axis of the sensor inserter prior to deployment of the physiological characteristic sensor and is removable from the adhesive patch by the sensor inserter upon deployment of the physiological characteristic sensor.

(10) Also provided according to various embodiment is a system for a physiological characteristic sensor deployed with a sensor inserter. The system includes an adhesive patch coupled to the physiological characteristic sensor. The adhesive patch is to couple the physiological characteristic sensor to an anatomy. The system includes a gravity resistance system coupled to the adhesive patch and to be coupled to the sensor inserter. The gravity resistance system maintains the adhesive

patch substantially perpendicular to a longitudinal axis of the sensor inserter prior to deployment of the physiological characteristic sensor and the gravity resistance system is to be removable from the sensor inserter by the adhesive patch upon deployment of the physiological characteristic sensor.

(11) Further provided is a system for a physiological characteristic sensor deployed with a sensor inserter. The system includes an adhesive patch coupled to the physiological characteristic sensor. The adhesive patch is to couple the physiological characteristic sensor to an anatomy. The system includes a gravity resistance system coupled to the adhesive patch and the sensor inserter. The gravity resistance system comprises a low tack adhesive paper that has a first surface positioned opposite a second surface by a fold. The first surface is coupled to the adhesive patch and the second surface is coupled to the sensor inserter. The gravity resistance system maintains the adhesive patch substantially perpendicular to a longitudinal axis of the sensor inserter prior to deployment of the physiological characteristic sensor and the gravity resistance system is removable from the sensor inserter by the adhesive patch upon deployment of the physiological characteristic sensor.

(12) This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the detailed description. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter. The details of one or more aspects of the disclosure are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of the techniques described in this disclosure will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) A more complete understanding of the subject matter may be derived by referring to the detailed description and claims when considered in conjunction with the following figures, wherein like reference numbers refer to similar elements throughout the figures.

(2) FIG. 1 is a perspective view of an exemplary sensor introduction system that includes a sensor inserter and a physiological characteristic sensor assembly having an exemplary gravity resistance system according to various teachings of the present disclosure;

(3) FIG. 2 is a cross-sectional view of the sensor introduction system of FIG. 1, taken along line 2-2 of FIG. 1;

(4) FIG. 3 is a top view of the physiological characteristic sensor assembly including the exemplary gravity resistance system of FIG. 1;

(5) FIG. 4 is a side view of the physiological characteristic sensor assembly including the exemplary gravity resistance system of FIG. 1;

(6) FIG. 5 is a cross-sectional view of another exemplary sensor introduction system that includes a sensor inserter and a physiological characteristic sensor assembly having an exemplary gravity resistance system according to various teachings of the present disclosure, taken from the perspective of line 2-2 of FIG. 1;

(7) FIG. 6 is a top view of the physiological characteristic sensor assembly including the exemplary gravity resistance system of FIG. 5;

(8) FIG. 7 is a cross-sectional view of the physiological characteristic sensor assembly including the exemplary gravity resistance system of FIG. 6, taken along line 7-7 of FIG. 6;

(9) FIG. 8 is a bottom view of the physiological characteristic sensor assembly including the exemplary gravity resistance system of FIG. 5, in which an adhesive patch associated with the physiological characteristic sensor assembly is removed for clarity;

- (10) FIG. 9 is a cross-sectional view of another exemplary sensor introduction system that includes a sensor inserter and a physiological characteristic sensor assembly having an exemplary gravity resistance system according to various teachings of the present disclosure, taken from the perspective of line 2-2 of FIG. 1;
- (11) FIG. 10 is a top view of the physiological characteristic sensor assembly including the exemplary gravity resistance system of FIG. 9;
- (12) FIG. 11 is a cross-sectional view of the physiological characteristic sensor assembly including the exemplary gravity resistance system of FIG. 10, taken along line 11-11 of FIG. 10;
- (13) FIG. 12 is a cross-sectional view of another exemplary sensor introduction system that includes a sensor inserter and a physiological characteristic sensor assembly having an exemplary gravity resistance system according to various teachings of the present disclosure, taken from the perspective of line 2-2 of FIG. 1;
- (14) FIG. 13 is a top view of the physiological characteristic sensor assembly including the exemplary gravity resistance system of FIG. 12;
- (15) FIG. 14 is a cross-sectional view of the physiological characteristic sensor assembly including the exemplary gravity resistance system of FIG. 13, taken along line 14-14 of FIG. 13; and
- (16) FIG. 15 is a cross-sectional view of another exemplary sensor introduction system that includes a sensor inserter and a physiological characteristic sensor assembly having an exemplary gravity resistance system according to various teachings of the present disclosure, taken from the perspective of line 2-2 of FIG. 1.

DETAILED DESCRIPTION

- (17) The following detailed description is merely illustrative in nature and is not intended to limit the embodiments of the subject matter or the application and uses of such embodiments. As used herein, the word “exemplary” means “serving as an example, instance, or illustration.” Any implementation described herein as exemplary is not necessarily to be construed as preferred or advantageous over other implementations. Furthermore, there is no intention to be bound by any expressed or implied theory presented in the preceding technical field, background, brief summary or the following detailed description.
- (18) Certain terminology may be used in the following description for the purpose of reference only, and thus are not intended to be limiting. For example, terms such as “top”, “bottom”, “upper”, “lower”, “above”, and “below” could be used to refer to directions in the drawings to which reference is made. Terms such as “front”, “back”, “rear”, “side”, “outboard”, and “inboard” could be used to describe the orientation and/or location of portions of the component within a consistent but arbitrary frame of reference which is made clear by reference to the text and the associated drawings describing the component under discussion. Such terminology may include the words specifically mentioned above, derivatives thereof, and words of similar import. Similarly, the terms “first”, “second”, and other such numerical terms referring to structures do not imply a sequence or order unless clearly indicated by the context.
- (19) As used herein, the term “axial” refers to a direction that is generally parallel to or coincident with an axis of rotation, axis of symmetry, or centerline of a component or components. For example, in a cylinder or disc with a centerline and generally circular ends or opposing faces, the “axial” direction may refer to the direction that generally extends in parallel to the centerline between the opposite ends or faces. In certain instances, the term “axial” may be utilized with respect to components that are not cylindrical (or otherwise radially symmetric). For example, the “axial” direction for a rectangular housing containing a rotating shaft may be viewed as a direction that is generally parallel to or coincident with the rotational axis of the shaft. Furthermore, the term “radially” as used herein may refer to a direction or a relationship of components with respect to a line extending outward from a shared centerline, axis, or similar reference, for example in a plane of a cylinder or disc that is perpendicular to the centerline or axis. In certain instances, components may be viewed as “radially” aligned even though one or both of the components may not be

cylindrical (or otherwise radially symmetric). Furthermore, the terms “axial” and “radial” (and any derivatives) may encompass directional relationships that are other than precisely aligned with (e.g., oblique to) the true axial and radial dimensions, provided the relationship is predominantly in the respective nominal axial or radial direction. As used herein, the term “transverse” denotes an axis that crosses another axis at an angle such that the axis and the other axis are neither substantially perpendicular nor substantially parallel.

(20) The following description relates to various embodiments of systems for adhesive skin patch gravity resistance. The systems described herein inhibit or mitigate the effects of gravity acting on an adhesive skin patch, during storage, for example, which ensures that the skin patch is properly adhered to a user. It should be noted that while the adhesive skin patch is described herein as being used with a glucose sensor, such as a glucose sensor associated with a continuous glucose monitor, it will be understood that the adhesive skin patch may be employed with a variety of other sensors, such as cardiac monitors, body temperature sensors, EKG monitors etc., medical devices, and/or other components that are intended to be affixed to the body of a user. Thus, while the non-limiting examples described below relate to a medical device used to treat diabetes (more specifically, an adhesive skin patch coupled to a continuous glucose monitor), embodiments of the disclosed subject matter are not so limited.

(21) Generally, the glucose sensor employed with the adhesive patch is a continuous glucose sensor of the type used by diabetic users. For the sake of brevity, conventional aspects and technology related to glucose sensors and glucose sensor fabrication may not be described in detail here. In this regard, known and/or conventional aspects of glucose sensors and their manufacturing may be of the type described in, but not limited to: U.S. Pat. Nos. 6,892,085, 7,468,033 and 9,295,786; and United States patent application number 2009/0299301 (which are each incorporated by reference herein).

(22) With reference to FIG. 1, FIG. 1 is a perspective view of a sensor introduction assembly **100**. In one example, the sensor introduction assembly **100** includes a physiological characteristic sensor assembly **102** and a sensor inserter **104**. It should be noted that in certain embodiments, the sensor inserter **104** and the physiological characteristic sensor **108** may comprise the insertion device and the sensor transmitter assembly described in commonly assigned U.S. Patent Publication No. 2017/0290533 to Antonio, et al., the relevant portion of which is incorporated herein by reference. In this example, with additional reference to FIG. 2, the physiological characteristic sensor assembly **102** includes a physiological characteristic sensor **108**, an adhesive skin patch or adhesive patch **110** and a gravity resistance system **112**. Generally, the components of the physiological characteristic sensor assembly **102** are coupled together as a single unit. The physiological characteristic sensor assembly **102** and the sensor inserter **104** may be packaged together for use by a consumer.

(23) Certain features, aspects, and characteristics of the sensor inserter **104**, the physiological characteristic sensor **108** and the adhesive patch **110** may be conventional and, as such, will not be described in detail here. Briefly, the physiological characteristic sensor **108** can be pre-connected as part of a sensor set, which could also include a sensor electronics module (not shown), such as a wireless transmitter that communicates with an infusion pump, a monitor device, or the like, which connects to the physiological characteristic sensor **108** after the insertion or deployment of a portion of the physiological characteristic sensor **108** in the body of the user. In one example, the physiological characteristic sensor **108** includes a glucose sensor **122** and a sensor base **124**. It should be noted that the physiological characteristic sensor **108** is not limited to a glucose sensor, but rather, various other physiological characteristic sensors may be employed. The glucose sensor **122** may be provided as an integral part of the sensor base **124**. The sensor base **124** gives structural support to the glucose sensor **122**, and facilitates entry of the glucose sensor **122** into the body of the user. The glucose sensor **122** is an electrochemical sensor that includes the glucose oxidase enzyme, as is well understood by those familiar with glucose sensor technology. The

glucose oxidase enzyme enables the glucose sensor **122** to monitor blood glucose levels in a diabetic patient or user by effecting a reaction of glucose and oxygen. Again, although certain embodiments pertain to glucose sensors, the technology described here can be adapted for use with any one of the wide variety of sensors known in the art. Generally, the glucose sensor **122** is positionable in subcutaneous tissue of the user by an insertion needle **126** of the sensor inserter **104** to measure the glucose oxidase enzyme.

(24) The sensor base **124** is coupled to the sensor inserter **104** and is coupled to the adhesive patch **110**. The sensor base **124** is removably coupled to the sensor inserter **104**. The sensor base **124** may also feature electrical and physical interfaces and elements that accommodate the sensor electronics module, such as the wireless transmitter that communicates with the infusion pump, the monitor device, or the like. In certain embodiments the sensor base **124** is composed at least in part from a plastic material. For the embodiment described here, the bulk of the sensor base **124** is formed as a molded plastic component. In one example, the sensor base **124** is formed from acrylonitrile butadiene styrene, nylon, an acrylonitrile butadiene styrene polycarbonate blend, polyvinyl chloride, polytetrafluoroethylene (PTFE), polypropylene, polyether ether ketone (PEEK), polycarbonate or the like.

(25) The adhesive patch **110** is coupled to the sensor base **124** and affixes the sensor base **124**, and thus, the glucose sensor **122**, to an anatomy, such as the skin of the user. The adhesive patch **110** is contained within the sensor inserter **104** during packaging and shipping, and is exposed to the force of gravity G . The adhesive patch **110** may be composed of a flexible and breathable material with one or more adhesive layers, such as cloth, a bandage-like material, and the like. For example, suitable materials could include polyurethane, polyethylene, polyester, polypropylene, polytetrafluoroethylene (PTFE), or other polymers, to which one or more adhesive layers are applied.

(26) The sensor inserter **104** is coupled to the physiological characteristic sensor **108** and is manipulatable by a user to couple the glucose sensor **122** to the user. With continued reference to FIG. 2, the sensor inserter **104** includes a housing **130**, a cradle or monitor support **132**, one or more biasing members or springs **134** and a lid or cover **136**. In one example, the housing **130** surrounds the physiological characteristic sensor assembly **102** and encloses the physiological characteristic sensor assembly **102** to enable sterilization of the physiological characteristic sensor assembly **102**, for example. The housing **130** may include one or more features, such as movable tabs, that cooperate with the monitor support **132** to deploy the physiological characteristic sensor **108** into the anatomy. The monitor support **132** is coupled to the physiological characteristic sensor **108**, and is movable relative to the housing **130** to deploy the physiological characteristic sensor **108** into the anatomy. For example, the application of a force to the housing **130** may bias the tabs to release the monitor support **132** to enable a spring **134** associated with the monitor support **132** to drive the monitor support **132** to deploy the physiological characteristic sensor **108** into the anatomy. Once released, another spring **134b** cooperates with the monitor support **132** to move a needle retractor **131** relative to the housing **130**. The cover **136** surrounds a circumferentially open end of the housing **130**, and encloses the housing **130**. Generally, the cover **136** is coupled to the housing **130** such that the adhesive patch **110** is unsupported by the cover **136**. As will be discussed, the gravity resistance system **112** inhibits or mitigates the force of gravity G from pulling down on the unsupported adhesive patch **110**, which in turn, inhibits or mitigates the drooping or sagging of the adhesive patch **110** within the sensor inserter **104** ensuring full contact is made between an entirety of the adhesive patch **110** and the anatomy of the user.

(27) In one example, with reference to FIG. 3, the gravity resistance system **112** is shown in greater detail. FIG. 3 is a top view of the physiological characteristic sensor assembly **102**, which illustrates the gravity resistance system **112** coupled to the adhesive patch **110**. In this example, the gravity resistance system **112** is a low tack adhesive cast paper, which is coupled to the adhesive patch **110** and the monitor support **132** (FIG. 2). The gravity resistance system **112** includes a first,

top surface **140** and a second, bottom surface **142**, which are interconnected at a fold **144** (FIG. 4). The gravity resistance system **112** is substantially annular, and defines an aperture **146**, which is sized to enable the gravity resistance system **112** to be positioned about a perimeter of the sensor base **124**. Generally, the gravity resistance system **112** surrounds an entirety of a circumference of the sensor base **124**, and may include a slit **148**. The slit **148** enables the removal of the gravity resistance system **112** from the adhesive patch **110**, if desired, by the user once the physiological characteristic sensor **108** is coupled to the anatomy. In this example, the slit **148** is defined at an end **150** of the gravity resistance system **112** that includes the fold **144**. The fold **144** may be configured such that the end **150** extends for a distance **D1**, which is different and less than a distance **D2** of an opposed end **152** of the gravity resistance system **112**. In this example, the gravity resistance system **112** is coupled to a surface **110a** of the adhesive patch **110** along a perimeter **110b** of the adhesive patch **110**, and extends for a distance **D3** from the perimeter of the adhesive patch **110** toward the sensor base **124**. Generally, the gravity resistance system **112** is spaced apart from the sensor base **124** by a fourth distance **D4**, which is different and less than the distance **D3**.

(28) In this example, with reference to FIG. 4, the gravity resistance system **112** is composed of a base layer **112a** to which a low tack adhesive **112b** is applied. Generally, the low tack adhesive **112b** is only applied to a single surface of the base layer **112a**, so that when folded, the top surface **140** and the bottom surface **142** include the low tack adhesive **112b**, but facing surfaces **154** remain uncoated with the low tack adhesive **112b**. In one example, the base layer **112a** is composed of paper, poly-coated paper, polymers such as polyester film or HDPE film, etc.; and the low tack adhesive **112b** is composed of silicone, acrylic, etc. The low tack adhesive **112b** may be cast, coated, painted or otherwise coupled to the base layer **112a**. The low tack adhesive **112b** along the bottom surface **142** is coupled or adhered to the surface **110a** of the adhesive patch **110**, while the low tack adhesive **112b** along the top surface **140** is coupled or adhered to a surface **132a** of the monitor support **132** (FIG. 2). As used herein, a “low tack” adhesive is an adhesive that has a bond weak enough to enable easy separation of the adhesive in its intended use (such as, separation of the liner from the adhesive patch either before or after insertion). As used herein, “high tack” adhesive is an adhesive in which the bond is intended to be permanent (i.e. no separation). For example, as used herein, a “low tack” adhesive has about 0.5 ounce per inch (oz/in.) to about 5 ounce per inch (oz/in.) peel force adhesion to stainless steel per ASTM D6862-11 Standard Test Method for 90 Degree Peel Resistance of Adhesives, and a “high tack” adhesive has greater than 5 ounce per inch (oz/in.) peel force adhesion to stainless steel per ASTM D6862-11 Standard Test Method for 90 Degree Peel Resistance of Adhesives.

(29) In one example, with the physiological characteristic sensor **108** assembled and coupled to the adhesive patch **110** and the gravity resistance system **112** formed, the low tack adhesive **112b** on the bottom surface **142** is coupled to the adhesive patch **110** so as to surround the sensor base **124**. The top surface **140** is folded at the fold **144** over the bottom surface **142**. With the physiological characteristic sensor assembly **102** assembled, and the springs **134** and the monitor support **132** coupled to the housing **130**, with reference to FIG. 3, the physiological characteristic sensor assembly **102** is coupled to the sensor inserter **104** such that the low tack adhesive **112b** is coupled to the surface **132a** of the monitor support **132**. With the physiological characteristic sensor assembly **102** coupled to the monitor support **132**, the cover **136** is coupled to the housing **130** to enclose the physiological characteristic sensor assembly **102**. The sensor inserter **104**, including the physiological characteristic sensor assembly **102**, may be sterilized and shipped to an end user.

(30) Once received, the user may remove the cover **136** to expose the physiological characteristic sensor assembly **102**. The user may manipulate the sensor inserter **104** to deploy the physiological characteristic sensor assembly **102** onto the user. Once deployed, the low tack adhesive **112b** on the top surface **140** enables the removal of the sensor inserter **104** from the physiological characteristic sensor assembly **102** without uncoupling the adhesive patch **110** from the user. With the sensor

inserter **104** uncoupled from the physiological characteristic sensor assembly **102** and the physiological characteristic sensor assembly **102** deployed on the user, the user may pull the top surface **140** of the gravity resistance system **112** to remove the gravity resistance system **112** from the adhesive patch **110**, if desired.

(31) By providing the low tack adhesive **112b** on the top surface **140**, the sensor inserter **104** is removable from the physiological characteristic sensor **108** upon deployment without removing the adhesive patch **110** from the user. Thus, the gravity resistance system **112** is removable from the sensor inserter **104** by the adhesive patch **110** upon deployment of the physiological characteristic sensor **108**. In addition, the low tack adhesive **112b** on the bottom surface **142** allows for the use of larger adhesive patches **110**, while inhibiting the drooping of the adhesive patch **110**. In this regard, the gravity resistance system **112** adds structure and rigidity to the portion of the adhesive patch **110** that extends beyond the sensor base **124** (FIG. 3). Stated another way, the gravity resistance system **112** maintains the adhesive patch **110** substantially perpendicular to a longitudinal axis LA1 of the sensor inserter **104**, which ensures the adhesive patch **110**, when deployed, is properly coupled to the user. The fold **144** also allows for removal of the gravity resistance system **112** by the user upon deployment, if desired.

(32) It should be noted that in other embodiments, the gravity resistance system **112** may be configured differently to inhibit or mitigate the effects of gravity on the adhesive patch **110**. For example, with reference to FIG. 5, a sensor introduction assembly **200** is shown. As the sensor introduction assembly **200** includes the same or similar components as the sensor introduction assembly **100** discussed with regard to FIGS. 1-4, the same reference numerals will be used to denote the same or similar components. FIG. 5 is a schematic cross-sectional view, taken from the perspective of line 2-2 of FIG. 1. In this example, the sensor introduction assembly **200** includes a physiological characteristic sensor assembly **202** and a sensor inserter **204**. In this example, the physiological characteristic sensor assembly **202** includes the physiological characteristic sensor **108**, the adhesive patch **110** and a gravity resistance system **212**. Generally, the components of the physiological characteristic sensor assembly **102** are coupled together as a single unit. The physiological characteristic sensor assembly **202** and the sensor inserter **204** may be packaged together for use by a consumer.

(33) The physiological characteristic sensor **108** includes the glucose sensor **122** and the sensor base **124**. Generally, the glucose sensor **122** is positionable in subcutaneous tissue of the user by an insertion needle of the sensor inserter **204** to measure the glucose oxidase enzyme. The sensor base **124** is coupled to the sensor inserter **204** and is coupled to the adhesive patch **110**. The sensor base **124** is removably coupled to the sensor inserter **204**. The adhesive patch **110** is coupled to the sensor base **124** and affixes the sensor base **124**, and thus, the glucose sensor **122**, to the skin of the user. The adhesive patch **110** is contained within the sensor inserter **204** during packaging and shipping, and is exposed to the force of gravity G.

(34) The sensor inserter **204** is coupled to the physiological characteristic sensor **108** and is manipulatable by a user to couple the glucose sensor **122** to the user. Briefly, the sensor inserter **204** includes a housing **230**, a monitor support **232** and a lid or cover **236**. In one example, the housing **230** surrounds the physiological characteristic sensor assembly **202** and encloses the physiological characteristic sensor assembly **202** to enable sterilization of the physiological characteristic sensor assembly **202**, for example. The housing **230** may include one or more features that cooperate with the monitor support **232** to deploy the physiological characteristic sensor **108** into the anatomy. The monitor support **232** is coupled to the physiological characteristic sensor **108**, and is manipulated by the user to deploy the physiological characteristic sensor **108**. The cover **236** surrounds a circumferentially open end of the housing **230**, and encloses the housing **230**. Generally, the cover **236** is coupled to the housing **230** such that the adhesive patch **110** is unsupported by the cover **236**. As will be discussed, the gravity resistance system **212** inhibits or mitigates the force of gravity G from pulling down on the unsupported adhesive patch **110**, which

in turn, inhibits or mitigates the drooping or sagging of the adhesive patch **110** within the sensor inserter **104** ensuring full contact is made between an entirety of the adhesive patch **110** and the anatomy of the user.

(35) In one example, with reference to FIG. **6**, the gravity resistance system **212** is shown in greater detail. FIG. **6** is a top view of the physiological characteristic sensor assembly **202**, which illustrates the gravity resistance system **212** coupled to the adhesive patch **110**. With reference to FIGS. **6** and **8**, the gravity resistance system **212** includes a first, top surface **240** and a second, bottom surface **242** (FIG. **8**). In FIG. **8**, the adhesive patch **110** is removed for clarity. The gravity resistance system **212** is substantially annular, and defines an aperture **246**, which is sized to enable the gravity resistance system **212** to be positioned about a perimeter of the sensor base **124**. Generally, the gravity resistance system **212** surrounds an entirety of a circumference of the sensor base **124**. In this example, with reference to FIG. **7**, the gravity resistance system **212** is coupled to the surface **110a** of the adhesive patch **110** along the perimeter **110b** of the adhesive patch **110**, and extends for a distance **D5** from the perimeter of the adhesive patch **110** toward the sensor base **124**. Generally, the gravity resistance system **212** is spaced apart from the sensor base **124** by a sixth distance **D6**, which is different and less than the distance **D5**.

(36) In this example, the gravity resistance system **212** is a double sided differential adhesive, which includes a high tack adhesive layer **250** coupled to a low tack adhesive layer **252**. The high tack adhesive layer **250** is coupled to the monitor support **232** (FIG. **5**), and the low tack adhesive layer **252** is coupled to the adhesive patch **110**. In this example, high tack adhesive **250a** is coupled to or formed on opposed sides of a base layer. The base layer is composed of paper, poly-coated paper, polymers such as polyester film or HDPE film, etc. The top surface **240** of the gravity resistance system **212** is defined by one side **250b** of the high tack adhesive layer **250**, which is coupled to or formed on the base layer. In one example, the high tack adhesive **250a** is composed of silicone, acrylic, etc. The high tack adhesive **250a** may be cast, coated, painted or otherwise coupled to the base layer. The opposed side **250c** of the high tack adhesive layer **250** formed on the base layer is coupled or adhered to the low tack adhesive layer **252**.

(37) In this example, low tack adhesive **252a** is coupled to or formed on opposed sides of a second base layer. The bottom surface **242** of the gravity resistance system **212** is defined by one side **252b** of the low tack adhesive layer **252**, which is coupled to or formed on the second base layer. The second base layer is composed of paper, poly-coated paper, polymers such as polyester film or HDPE film. In one example, the low tack adhesive **252a** is composed of silicone, acrylic, etc. The low tack adhesive **252a** may be cast, coated, painted or otherwise coupled to the second base layer. The opposed side **252c** of low tack adhesive layer **252** formed on the second base layer is coupled or adhered to the side **250c** of the high tack adhesive layer **250** to form the gravity resistance system **212**. Thus, the high tack adhesive **250a** is a first tack adhesive, and the low tack adhesive **252a** is a second tack adhesive, with the second tack adhesive different and less than the first tack adhesive. It should be noted that for ease of illustration, the base layer and the second base layer are not shown in the drawings as these paper or film layers have a predetermined nominal thickness.

(38) In one example, with the physiological characteristic sensor **108** assembled and coupled to the adhesive patch **110** and the gravity resistance system **212** formed, with reference to FIG. **5**, the low tack adhesive layer **252** on the bottom surface **242** is coupled to the adhesive patch **110** so as to surround the sensor base **124**. With the physiological characteristic sensor assembly **202** assembled and the monitor support **232** coupled to the housing **230**, the physiological characteristic sensor assembly **202** is coupled to the sensor inserter **204** such that the high tack adhesive layer **250** is coupled to the surface **232a** of the monitor support **232**. With the physiological characteristic sensor assembly **202** coupled to the monitor support **232**, the cover **236** is coupled to the housing **230** to enclose the physiological characteristic sensor assembly **202**. The sensor inserter **204**, including the physiological characteristic sensor assembly **202**, may be sterilized and shipped to an end user.

(39) Once received, the user may remove the cover **236** to expose the physiological characteristic sensor assembly **202**. The user may manipulate the sensor inserter **204** to deploy the physiological characteristic sensor assembly **202** onto the user. Once deployed, the high tack adhesive layer **250** on the top surface **240** retains the gravity resistance system **212** on the sensor inserter **204**, and the low tack adhesive layer **252** enables the removal of the gravity resistance system **212** from the adhesive patch **110** without uncoupling the adhesive patch **110** from the user. Thus, the gravity resistance system **212** is removable from the adhesive patch **110** by the sensor inserter **204** upon deployment of the physiological characteristic sensor **108**. The differential adhesive of the gravity resistance system **212** enables the sensor inserter **204** to be uncoupled from the physiological characteristic sensor **108** when the physiological characteristic sensor **108** is coupled to the user with the adhesive patch **110** without uncoupling the physiological characteristic sensor **108** and the adhesive patch **110** from the user.

(40) By providing the high tack adhesive layer **250** on the top surface **240** and the low tack adhesive layer **252** on the bottom surface **242**, the gravity resistance system **212** is retained on the sensor inserter **204** and is removable from the physiological characteristic sensor **108** upon deployment without removing the adhesive patch **110** from the user. In addition, the low tack adhesive layer **252** on the bottom surface **242** allows for the use of larger adhesive patches **110**, while inhibiting the drooping of the adhesive patch **110**. In this regard, the gravity resistance system **212** adds structure and rigidity to the portion of the adhesive patch **110** that extends beyond the sensor base **124**. Stated another way, the gravity resistance system **212** maintains the adhesive patch **110** substantially perpendicular to a longitudinal axis LA2 of the sensor inserter **204**, which ensures the adhesive patch **110**, when deployed, is properly coupled to the user.

(41) It should be noted that in other embodiments, the gravity resistance system **112** may be configured differently to inhibit or mitigate the effects of gravity on the adhesive patch **110**. For example, with reference to FIG. 9, a sensor introduction assembly **300** is shown. As the sensor introduction assembly **300** includes the same or similar components as the sensor introduction assembly **100** discussed with regard to FIGS. 1-4 and the sensor introduction assembly **200** discussed with regard to FIGS. 5-8, the same reference numerals will be used to denote the same or similar components. FIG. 9 is a schematic cross-sectional view, taken from the perspective of line 2-2 of FIG. 1. In this example, the sensor introduction assembly **300** includes a physiological characteristic sensor assembly **302** and the sensor inserter **204**. In this example, the physiological characteristic sensor assembly **302** includes the physiological characteristic sensor **108**, the adhesive patch **110** and a gravity resistance system **312**. Generally, the components of the physiological characteristic sensor assembly **302** are coupled together as a single unit. The physiological characteristic sensor assembly **302** and the sensor inserter **204** may be packaged together for use by a consumer.

(42) The physiological characteristic sensor **108** includes the glucose sensor **122** and the sensor base **124**. The sensor base **124** is coupled to the sensor inserter **204** and is coupled to the adhesive patch **110**. The sensor base **124** is removably coupled to the sensor inserter **204**. The adhesive patch **110** is coupled to the sensor base **124** and affixes the sensor base **124**, and thus, the glucose sensor **122**, to the skin of the user. The adhesive patch **110** is contained within the sensor inserter **204** during packaging and shipping, and is exposed to the force of gravity G.

(43) The sensor inserter **204** is coupled to the physiological characteristic sensor **108** and is manipulatable by a user to couple the glucose sensor **122** to the user. Briefly, the sensor inserter **204** includes the housing **230**, the monitor support **232** and the lid or cover **236**. In one example, the housing **230** surrounds the physiological characteristic sensor assembly **302** and encloses the physiological characteristic sensor assembly **302** to enable sterilization of the physiological characteristic sensor assembly **302**, for example. The housing **230** may include one or more features that cooperate with the monitor support **232** to deploy the physiological characteristic sensor **108** into the anatomy. The monitor support **232** is coupled to the physiological characteristic

sensor **108**, and is manipulated by the user to deploy the physiological characteristic sensor **108**. The cover **236** surrounds the circumferentially open end of the housing **230**, and encloses the housing **230**. Generally, the cover **236** is coupled to the housing **230** such that the adhesive patch **110** is unsupported by the cover **236**. As will be discussed, the gravity resistance system **312** inhibits or mitigates the force of gravity G from pulling down on the unsupported adhesive patch **110**, which in turn, inhibits or mitigates the drooping or sagging of the adhesive patch **110** within the sensor inserter **204** ensuring full contact is made between an entirety of the adhesive patch **110** and the anatomy of the user.

(44) In one example, with reference to FIG. **10**, the gravity resistance system **312** is shown in greater detail. FIG. **10** is a top view of the physiological characteristic sensor assembly **302**, which illustrates the gravity resistance system **312** coupled to the adhesive patch **110**. With reference to FIGS. **10** and **11**, the gravity resistance system **312** includes a first, top surface **340** and a second, bottom surface **342** (FIG. **11**). The gravity resistance system **312** is substantially annular, and defines an aperture **346**, which is sized to enable the gravity resistance system **312** to be positioned about a perimeter of the sensor base **124** (FIG. **10**). Generally, the gravity resistance system **312** surrounds an entirety of a circumference of the sensor base **124**. In this example, with reference to FIG. **11**, the gravity resistance system **312** is coupled to the surface **110a** of the adhesive patch **110** along the perimeter **110b** of the adhesive patch **110**, and extends for a distance $D7$ from the perimeter of the adhesive patch **110** toward the sensor base **124**. Generally, the gravity resistance system **312** is spaced apart from the sensor base **124** by an eighth distance $D8$, which is different and less than the distance $D7$.

(45) In this example, the gravity resistance system **312** is a single layer double sided differential adhesive, which includes a high tack adhesive **350** on a first side **312a** and a low tack adhesive **352** on a second side **312b**. The high tack adhesive **350** is coupled to the monitor support **332** (FIG. **9**), and the low tack adhesive **352** is coupled to the adhesive patch **110**. In this example, high tack adhesive **350** and the low tack adhesive **352** are each coupled to or formed on opposed sides of a base layer. The base layer is composed of paper, poly-coated paper, polymers such as polyester film or HDPE film. The top surface **340** of the gravity resistance system **312** is defined by the high tack adhesive **350** and the bottom surface **342** of the gravity resistance system **312** is defined by the low tack adhesive **352**, which are coupled to or formed on opposed sides of the base layer. In one example, the high tack adhesive **350** is composed of synthetic rubber adhesives, acrylic, etc. The high tack adhesive **350** may be cast, coated, painted or otherwise coupled to the base layer. The low tack adhesive **352** is coupled to or formed on a second, opposed side of the base layer. In one example, the low tack adhesive **352** is composed of silicone, acrylic, etc. The low tack adhesive **352** may be cast, coated, painted or otherwise coupled to the base layer. It should be noted that for ease of illustration, the base layer is not shown in the drawings as this paper or film layer has a predetermined nominal thickness.

(46) In one example, with the physiological characteristic sensor **108** assembled and coupled to the adhesive patch **110** and the gravity resistance system **312** formed, with reference to FIG. **9**, the low tack adhesive **352** on the bottom surface **342** is coupled to the adhesive patch **110** so as to surround the sensor base **124**. With the physiological characteristic sensor assembly **302** assembled and the monitor support **232** coupled to the housing **230**, the physiological characteristic sensor assembly **302** is coupled to the sensor inserter **204** such that the high tack adhesive **350** is coupled to the surface **232a** of the monitor support **232**. With the physiological characteristic sensor assembly **302** coupled to the monitor support **232**, the cover **236** is coupled to the housing **230** to enclose the physiological characteristic sensor assembly **302**. The sensor inserter **204**, including the physiological characteristic sensor assembly **302**, may be sterilized and shipped to an end user.

(47) Once received, the user may remove the cover **236** to expose the physiological characteristic sensor assembly **302**. The user may manipulate the sensor inserter **204** to deploy the physiological characteristic sensor assembly **302** onto the user. Once deployed, the high tack adhesive **350** on the

top surface **340** retains the gravity resistance system **312** on the sensor inserter **204**, and the low tack adhesive **352** enables the removal of the gravity resistance system **312** from the adhesive patch **110** without uncoupling the adhesive patch **110** from the user. Thus, the differential adhesive of the gravity resistance system **312** enables the sensor inserter **204** to be uncoupled from the physiological characteristic sensor **108** when the physiological characteristic sensor **108** is coupled to the user with the adhesive patch **110** without uncoupling the physiological characteristic sensor **108** and the adhesive patch **110** from the user.

(48) By providing the high tack adhesive **350** on the top surface **340** and the low tack adhesive **352** on the bottom surface **342**, the gravity resistance system **312** is retained on the sensor inserter **204** and is removable from the physiological characteristic sensor **108** upon deployment without removing the adhesive patch **110** from the user. Thus, the gravity resistance system **312** is removable from the adhesive patch **110** by the sensor inserter **204** upon deployment of the physiological characteristic sensor **108**. In addition, the low tack adhesive **352** on the bottom surface **342** allows for the use of larger adhesive patches **110**, while inhibiting the drooping of the adhesive patch **110**. In this regard, the gravity resistance system **312** adds structure and rigidity to the portion of the adhesive patch **110** that extends beyond the sensor base **124**. Stated another way, the gravity resistance system **312** maintains the adhesive patch **110** substantially perpendicular to the longitudinal axis LA2 of the sensor inserter **204**, which ensures the adhesive patch **110**, when deployed, is properly coupled to the user.

(49) It should be noted that in other embodiments, the gravity resistance system **112** may be configured differently to inhibit or mitigate the effects of gravity on the adhesive patch **110**. For example, with reference to FIG. **12**, a sensor introduction assembly **400** is shown. As the sensor introduction assembly **400** includes the same or similar components as the sensor introduction assembly **100** discussed with regard to FIGS. **1-4** and the sensor introduction assembly **200** discussed with regard to FIGS. **5-8**, the same reference numerals will be used to denote the same or similar components. FIG. **12** is a schematic cross-sectional view, taken from the perspective of line **2-2** of FIG. **1**. In this example, the sensor introduction assembly **400** includes a physiological characteristic sensor assembly **402** and the sensor inserter **204**. In this example, the physiological characteristic sensor assembly **402** includes the physiological characteristic sensor **108**, the adhesive patch **110** and a gravity resistance system **412**. Generally, the components of the physiological characteristic sensor assembly **402** are coupled together as a single unit. The physiological characteristic sensor assembly **402** and the sensor inserter **204** may be packaged together for use by a consumer.

(50) The physiological characteristic sensor **108** includes the glucose sensor **122** and the sensor base **124**. The sensor base **124** is coupled to the sensor inserter **204** and is coupled to the adhesive patch **110**. The sensor base **124** is removably coupled to the sensor inserter **204**. The adhesive patch **110** is coupled to the sensor base **124** and affixes the sensor base **124**, and thus, the glucose sensor **122**, to the skin of the user. The adhesive patch **110** is contained within the sensor inserter **204** during packaging and shipping, and is exposed to the force of gravity **G**.

(51) The sensor inserter **204** is coupled to the physiological characteristic sensor **108** and is manipulatable by a user to couple the glucose sensor **122** to the user. Briefly, the sensor inserter **204** includes the housing **230**, the monitor support **232** and the lid or cover **236**. In one example, the housing **230** surrounds the physiological characteristic sensor assembly **202** and encloses the physiological characteristic sensor assembly **202** to enable sterilization of the physiological characteristic sensor assembly **202**, for example. The housing **230** may include one or more features that cooperate with the monitor support **232** to deploy the physiological characteristic sensor **108** into the anatomy. The monitor support **232** is coupled to the physiological characteristic sensor **108**, and is manipulated by the user to deploy the physiological characteristic sensor **108** into the anatomy. The cover **236** surrounds the circumferentially open end of the housing **230**, and encloses the housing **230**. Generally, the cover **236** is coupled to the housing **230** such that the

adhesive patch **110** is unsupported by the cover **236**. As will be discussed, the gravity resistance system **412** inhibits or mitigates the force of gravity G from pulling down on the unsupported adhesive patch **110**, which in turn, inhibits or mitigates the drooping or sagging of the adhesive patch **110** within the sensor inserter **204** ensuring full contact is made between an entirety of the adhesive patch **110** and the anatomy of the user.

(52) In one example, with reference to FIG. **13**, the gravity resistance system **412** is shown in greater detail. FIG. **13** is a top view of the physiological characteristic sensor assembly **402**, which illustrates the gravity resistance system **412** coupled to the adhesive patch **110**. In this example, the gravity resistance system **412** includes a plurality of adhesive strips **414**, which are spaced apart about a perimeter of the sensor base **124**. The plurality of adhesive strips **414** is also spaced apart about a perimeter or circumference of the adhesive patch **110**. In this example, the gravity resistance system **412** includes four adhesive strips **414**, but it should be understood that the gravity resistance system **412** may include any number of adhesive strips **414**.

(53) Each of the adhesive strips **414** includes a first, top surface **440** and a second, bottom surface **442** (FIG. **14**). Each of the adhesive strips **414** is rectangular, with rounded edges. It should be noted, however, that the adhesive strips **414** may have any desired shape, and further, one or more of the adhesive strips **414** may have a different shape. In this example, each of the adhesive strips **414** has a length $L1$ and a width $W1$. The length $L1$ and width $W1$ are each predefined to ensure that the adhesive strips **414** provide rigidity to the adhesive patch **110**, while also ensuring that the adhesive strips **414** do not interfere with the removal of the adhesive patch **110** from the sensor inserter **204**, as will be discussed below. In one example, the length $L1$ is about 100 micrometers (μm) to about 1.0 millimeters (mm); and the width $W1$ is about 100 micrometers (μm) to about 5.0 millimeters (mm). Generally, the adhesive strips **414** are sized and located to interface with the monitor support **232**. Each of the adhesive strips **414** is positioned a distance $D9$ from the perimeter **110b** of the adhesive patch **110**, and distance $D10$ from a perimeter of the sensor base **124**. In one example, the distance $D9$ is about equal to or the same as the distance $D10$, and is about 0 millimeters (mm) to about 10 millimeters (mm).

(54) In this example, each of the adhesive strips **414** comprises a differential double sided adhesive, which includes a high tack adhesive **450** on the top surface **440** and a low tack adhesive **452** on the bottom surface **442**. The high tack adhesive **450** on the top surface **440** is coupled to the monitor support **232** (FIG. **12**), and the low tack adhesive **452** on the bottom surface **442** is coupled to the adhesive patch **110**. The top surface **440** of the gravity resistance system **412** is defined by the high tack adhesive **450** and the bottom surface **442** of the gravity resistance system **412** is defined by the low tack adhesive **452**. In this example, high tack adhesive **450** and the low tack adhesive **452** are each coupled to or formed on opposed sides of a base layer. The base layer is composed of paper, poly-coated paper, polymers such as polyester film or HDPE film. In one example, the high tack adhesive **450** is composed of synthetic rubber adhesive, acrylic, etc. The high tack adhesive **450** may be cast, coated, painted or otherwise coupled to the base layer. The low tack adhesive **452** is coupled to or formed on a second, opposed side of the base layer. In one example, the low tack adhesive **452** is composed of silicone, acrylic, etc. The low tack adhesive **452** may be cast, coated, painted or otherwise coupled to the base layer. It should be noted that for ease of illustration, the base layer is not shown in the drawings as this paper or film layer has a predetermined nominal thickness.

(55) In one example, with the physiological characteristic sensor **108** assembled and coupled to the adhesive patch **110** and the gravity resistance system **412** formed, with reference to FIG. **5**, the adhesive strips **414** are coupled to the adhesive patch **110** (via the low tack adhesive **452** on the bottom surface **442**) so as to be spaced apart about the perimeter **110b** of the adhesive patch **110**. With the physiological characteristic sensor assembly **402** assembled and the monitor support **232** coupled to the housing **230**, the physiological characteristic sensor assembly **402** is coupled to the sensor inserter **204** such that the high tack adhesive **450** on the top surface **440** is coupled to the

surface **232a** of the monitor support **232**. With the physiological characteristic sensor assembly **402** coupled to the monitor support **232**, the cover **236** is coupled to the housing **230** to enclose the physiological characteristic sensor assembly **402**. The sensor inserter **204**, including the physiological characteristic sensor assembly **402**, may be sterilized and shipped to an end user. (56) Once received, the user may remove the cover **236** to expose the physiological characteristic sensor assembly **402**. The user may manipulate the sensor inserter **204** to deploy the physiological characteristic sensor assembly **402** onto the user. Once deployed, the high tack adhesive **450** on the top surface **440** retains the gravity resistance system **412** on the sensor inserter **204**. Thus, the gravity resistance system **412** is removable from the adhesive patch **110** by the sensor inserter **204** upon deployment of the physiological characteristic sensor **108**. The gravity resistance system **412** enables the sensor inserter **204** to be uncoupled from the physiological characteristic sensor **108** when the physiological characteristic sensor **108** is coupled to the user with the adhesive patch **110** without uncoupling the physiological characteristic sensor **108** and the adhesive patch **110** from the user.

(57) By providing the adhesive strips **414** with the high tack adhesive layer **250** on the top surface **440** and the low tack adhesive **452** on the bottom surface **442**, the gravity resistance system **412** is retained on the sensor inserter **204** and is removable from the physiological characteristic sensor **108** upon deployment without removing the adhesive patch **110** from the user. In addition, the gravity resistance system **412** allows for the use of larger adhesive patches **110**, while inhibiting the drooping of the adhesive patch **110**. In this regard, the gravity resistance system **412** adds structure and rigidity to the portion of the adhesive patch **110** that extends beyond the sensor base **124**. Stated another way, the gravity resistance system **412** maintains the adhesive patch **110** substantially perpendicular to the longitudinal axis LA2 of the sensor inserter **204**, which ensures the adhesive patch **110**, when deployed, is properly coupled to the user.

(58) It should be noted that in other embodiments, the gravity resistance system **112** may be configured differently to inhibit or mitigate the effects of gravity on the adhesive patch **110**. For example, with reference to FIG. 15, a sensor introduction assembly **500** is shown. As the sensor introduction assembly **500** includes the same or similar components as the sensor introduction assembly **100** discussed with regard to FIGS. 1-4 and the sensor introduction assembly **200** discussed with regard to FIGS. 5-8, the same reference numerals will be used to denote the same or similar components. FIG. 15 is a schematic cross-sectional view, taken from the perspective of line 2-2 of FIG. 1. In this example, the sensor introduction assembly **500** includes a physiological characteristic sensor assembly **502** and a sensor inserter **504**. In this example, the physiological characteristic sensor assembly **502** includes the physiological characteristic sensor **108**, an adhesive skin patch or adhesive patch **510** and a gravity resistance system **512**. Generally, the components of the physiological characteristic sensor assembly **502** are coupled together as a single unit. The physiological characteristic sensor assembly **502** and the sensor inserter **504** may be packaged together for use by a consumer.

(59) The physiological characteristic sensor **108** includes the glucose sensor **122** and the sensor base **124**. Generally, the glucose sensor **122** is positionable in subcutaneous tissue of the user by an insertion needle of the sensor inserter **504** to measure the glucose oxidase enzyme. The sensor base **124** is coupled to the sensor inserter **504** and is coupled to the adhesive patch **110**. The sensor base **124** is removably coupled to the sensor inserter **204**.

(60) The adhesive patch **510** is coupled to the sensor base **124** and affixes the sensor base **124**, and thus, the glucose sensor **122**, to the skin of the user. The adhesive patch **510** is contained within the sensor inserter **504** during packaging and shipping, and is exposed to the force of gravity G. The adhesive patch **510** may be composed of a flexible and breathable material with one or more adhesive layers, such as cloth, a bandage-like material, and the like. For example, suitable materials could include polyurethane, polyethylene, polyester, polypropylene, polytetrafluoroethylene (PTFE), or other polymers, to which one or more adhesive layers are applied. In this example, the

adhesive patch **510** includes an electrically charged surface or first charged surface **510a**. The first charged surface **510a** is opposite the surface **510b**, which is coupled to the user. In one example, the first charged surface **510a** has a positive electric charge, which cooperates with a negatively charged surface of the sensor inserter **504**, as will be discussed. In other examples, the first charged surface **510a** may have a negatively charged surface, which cooperates with a corresponding positively charged surface of the sensor inserter **504**. The first charged surface **510a** may be charged using contact-induced charge separation, charge-induced charge separation, etc. For contact-induced charge separation, the amount of charge applied, and the polarity of the charge depends on the materials and surface roughness.

(61) The sensor inserter **504** is coupled to the physiological characteristic sensor **108** and is manipulatable by a user to couple the glucose sensor **122** to the user. Briefly, the sensor inserter **504** includes the housing **230**, a monitor support **532** and the lid or cover **236**. In one example, the housing **230** surrounds the physiological characteristic sensor assembly **502** and encloses the physiological characteristic sensor assembly **502** to enable sterilization of the physiological characteristic sensor assembly **502**, for example. The housing **230** may include one or more features that cooperate with the monitor support **532** to deploy the physiological characteristic sensor **108** into the anatomy. The monitor support **532** is coupled to the physiological characteristic sensor **108**, and is manipulated by the user to deploy the physiological characteristic sensor **108** into the anatomy. In this example, the monitor support **532** includes an electrically charged surface or second charged surface **532a**. The second charged surface **532a** faces the adhesive patch **110**. In one example, the second charged surface **532a** has a negative electric charge, which cooperates with the first charged surface **510a** of the adhesive patch **510**. The second charged surface **532a** may be charged using contact-induced charge separation, charge-induced charge separation, etc. For contact-induced charge separation, the amount of charge applied, and the polarity of the charge depends on the materials and surface roughness.

(62) In the example of contact-induced charge separation, the first charged surface **510a** of the adhesive patch **510** is composed of a material that is more negatively charged in the triboelectric series, such as a polyurethane film. The second charged surface **532a** of the monitor support **532** is composed of a material that is more positively charged in the triboelectric series than the material of the first charged surface **510a** of the adhesive patch **510**, such as a nylon. The contact between the first charged surface **510a** and the second charged surface **532a** results in adhesion between the two surfaces **510a**, **532a** as the electrons are exchanged and are attracted to one another by opposite charge build up on each surface, which inhibits the drooping of the adhesive patch **510**. It should be noted that the materials selected herein are merely examples, as any materials that are separated along the triboelectric series relative to one another may be used for the adhesive patch **510** and the monitor support **532** so long as the contact between the first charged surface **510a** and the second charged surface **532a** results in adhesion between the two surfaces **510a**, **532a** due to electron exchange and attraction due to opposite charge build up on the respective surfaces **510a**, **532a**. It should be noted that an entirety of the monitor support **532** may be composed of the predetermined material, or merely a surface of the monitor support **532**, such as the second charged surface **532a**, may be formed of the predetermined material.

(63) In the example of charge-induced charge separation, the first charged surface **510a** may be initially composed of an electrically neutral material such as polyester that has been electrically grounded to have a net neutral charge. A negatively charged object may be brought near the first charged surface **510a**, to induce a positive charge on the first charged surface **510a** as the positive electrons associated with the first charged surface **510a** move toward the negatively charged object. Similarly, the second charged surface **532a** may be composed of an electrically neutral material such as polycarbonate that has been electrically grounded to have a net neutral charge. A positively charged object may be brought near the second charged surface **532a**, to induce a negative charge on the second charged surface **532a** as the negative electrons associated with the first charged

surface **510a** move toward the positively charged object. When the physiological characteristic sensor **108** is coupled to the sensor inserter **504**, the negatively-charged second charged surface **532a** attracts the positively-charged first charged surface **510a**, which inhibits the drooping of the adhesive patch **510**.

(64) The cover **236** surrounds the circumferentially open end of the housing **230**, and encloses the housing **230**. Generally, the cover **236** is coupled to the housing **230** such that the adhesive patch **510** is unsupported by the cover **236**. As discussed, the gravity resistance system **512** inhibits or mitigates the force of gravity G from pulling down on the unsupported adhesive patch **510**, which in turn, inhibits or mitigates the drooping or sagging of the adhesive patch **510** within the sensor inserter **204** ensuring full contact is made between an entirety of the adhesive patch **510** and the anatomy of the user.

(65) In one example, with the physiological characteristic sensor **108** assembled and coupled to the adhesive patch **510** and the gravity resistance system **512** formed, the first charged surface **510a** is charged to have the respective electric charge, in this example, a positive electric charge. The second charged surface **532a** of the monitor support **532** is charged to have the respective electric charge, in this example, a negative electric charge. With the physiological characteristic sensor assembly **502** assembled and the monitor support **532** coupled to the housing **230**, the physiological characteristic sensor assembly **502** is coupled to the sensor inserter **504** such that the first charged surface **510a** of the adhesive patch **510** is electrically attracted to the second charged surface **532a** of the monitor support **532**. With the physiological characteristic sensor assembly **502** coupled to the monitor support **532**, the cover **236** is coupled to the housing **230** to enclose the physiological characteristic sensor assembly **502**. The sensor inserter **504**, including the physiological characteristic sensor assembly **502**, may be sterilized and shipped to an end user.

(66) Once received, the user may remove the cover **236** to expose the physiological characteristic sensor assembly **502**. The user may manipulate the sensor inserter **504** to deploy the physiological characteristic sensor assembly **502** onto the user. The weak attractive force between the first charged surface **510a** and the second charged surface **532a** enables the sensor inserter **504** to be removed from the physiological characteristic sensor assembly **502**. Thus, the gravity resistance system **512** enables the sensor inserter **504** to be uncoupled from the physiological characteristic sensor **108** when the physiological characteristic sensor **108** is coupled to the user with the adhesive patch **510** without uncoupling the physiological characteristic sensor **108** and the adhesive patch **510** from the user. Further, by providing the attractive force between the first charged surface **510a** and the second charged surface **532a**, the gravity resistance system **512** allows for the use of larger adhesive patches **110**, while inhibiting the drooping of the adhesive patch **510**. In this regard, the attractive force between the first charged surface **510a** and the second charged surface **532a** maintains the adhesive patch **510** substantially perpendicular to a longitudinal axis **LA5** of the sensor inserter **504**, which ensures the adhesive patch **510**, when deployed, is properly coupled to the user. Stated another way, the adhesive patch **510** includes a first electrically charged surface or the first charged surface **510a** having a first electric charge, which in this example is a positive electric charge, the sensor inserter **504** includes a second electrically charged surface or the second charged surface **532a** having a second electric charge, which in this example is a negative electric charge, and the first electric charge is different than the second electric charge to maintain the adhesive patch **510** substantially perpendicular to the longitudinal axis **LA5** of the sensor inserter **504**.

(67) It should be noted that the sensor inserter **104**, **204**, **504** described and illustrated herein is merely exemplary, as any device may be employed with the gravity resistance system **112**, **212**, **312**, **412**, **512** to deploy the physiological characteristic sensor **108** into the anatomy. For example, an exemplary sensor inserter may include merely a monitor support, such as the monitor support **232**, **532**, which is manually manipulated by a user to deploy the physiological characteristic sensor **108** into the anatomy. Moreover, it should be noted that the sensor inserter **204**, **504** may comprise

the sensor inserter **104** discussed with regard to FIGS. **1-4** or the insertion device described in commonly assigned U.S. Patent Publication No. 2017/0290533 to Antonio, et al., the relevant portion of which was previously incorporated herein by reference.

(68) While at least one exemplary embodiment has been presented in the foregoing detailed description, it should be appreciated that a vast number of variations exist. It should also be appreciated that the exemplary embodiment or embodiments described herein are not intended to limit the scope, applicability, or configuration of the claimed subject matter in any way. Rather, the foregoing detailed description will provide those skilled in the art with a convenient road map for implementing the described embodiment or embodiments. It should be understood that various changes can be made in the function and arrangement of elements without departing from the scope defined by the claims, which includes known equivalents and foreseeable equivalents at the time of filing this patent application.

(69) It should be understood that various aspects disclosed herein may be combined in different combinations than the combinations specifically presented in the description and accompanying drawings. It should also be understood that, depending on the example, certain acts or events of any of the processes or methods described herein may be performed in a different sequence, may be added, merged, or left out altogether (e.g., all described acts or events may not be necessary to carry out the techniques). In addition, while certain aspects of this disclosure are described as being performed by a single module or unit for purposes of clarity, it should be understood that the techniques of this disclosure may be performed by a combination of units or modules associated with, for example, a medical device.

Claims

1. A system comprising: A physiological sensor having a base; an inserter including a support coupled to the physiological sensor, the support being movable to deploy the physiological sensor; an adhesive patch coupled to the base of the physiological sensor, the adhesive patch having a first surface and a second surface opposite the first surface, the second surface configured to couple the physiological sensor to an anatomy upon deployment; and a gravity resistance system including a first tack adhesive coupled to the support of the inserter and a second tack adhesive coupled to the first surface of the adhesive patch and being spaced apart from the base, the second tack adhesive being less tacky than the first tack adhesive, the gravity resistance system maintains at least a peripheral portion of the adhesive patch adhered to the support prior to deployment of the physiological sensor and is removable from the adhesive patch upon deployment of the physiological sensor.
2. The system of claim 1, wherein the second tack adhesive of the gravity resistance system extends at least partially around a perimeter of the base of the physiological sensor.
3. The system of claim 1, wherein the second tack adhesive of the gravity resistance system extends completely around a perimeter of the base of the physiological sensor.
4. The system of claim 1, wherein the gravity resistance system comprises a double sided adhesive such that the first tack adhesive is opposite the second tack adhesive.
5. The system of claim 1, wherein the gravity resistance system comprises a plurality of adhesive strips, which are spaced apart about a perimeter of the adhesive patch.
6. The system of claim 1, wherein the gravity resistance system comprises a plurality of adhesive strips, each adhesive strip of the plurality of adhesive strips including the first tack adhesive opposite the second tack adhesive, which are spaced apart about a perimeter of the adhesive patch and the base is coupled to the first surface of the adhesive patch.
7. The system of claim 1, wherein the gravity resistance system comprises a first adhesive layer with the first tack adhesive on opposed sides, and a second adhesive layer with the second tack adhesive on opposed sides.

8. The system of claim 1, wherein the physiological sensor includes an insertable member, the insertable member extending through the base and insertable into the anatomy upon deployment of the physiological sensor, wherein the gravity resistance system extends at least partially around the insertable member.
9. A system comprising: a physiological sensor having a base; an adhesive patch coupled to the base of the physiological sensor, the adhesive patch having a first surface and a second surface opposite the first surface, the second surface configured to couple the physiological sensor to an anatomy upon deployment; and a gravity resistance system including a first tack adhesive configured to adhere to a support of an inserter prior to deployment of the physiological sensor, and a second tack adhesive adhered to the first surface of the adhesive patch and being spaced apart from the base, the second tack adhesive being less tacky than the first tack adhesive, the gravity resistance system being configured to maintain at least a peripheral portion of the adhesive patch adhered to the support prior to deployment of the physiological sensor and is removable from the adhesive patch upon deployment of the physiological sensor.
10. The system of claim 9, wherein the second tack adhesive of the gravity resistance system extends at least partially around a perimeter of the base of the physiological sensor.
11. The system of claim 9, wherein the second tack adhesive of the gravity resistance system extends completely around a perimeter of the base of the physiological sensor.
12. The system of claim 9, wherein the gravity resistance system comprises a double sided adhesive such that the first tack adhesive is opposite the second tack adhesive.
13. The system of claim 9, wherein the gravity resistance system comprises a plurality of adhesive strips, which are spaced apart about a perimeter of the adhesive patch.
14. The system of claim 9, wherein the gravity resistance system comprises a plurality of adhesive strips, each adhesive strip of the plurality of adhesive strips including the first tack adhesive opposite the second tack adhesive, which are spaced apart about a perimeter of the adhesive patch and the base is coupled to the first surface of the adhesive patch.
15. The system of claim 9, wherein the gravity resistance system comprises a first adhesive layer with the first tack adhesive on opposed sides, and a second adhesive layer with the second tack adhesive on opposed sides.
16. The system of claim 9, wherein the physiological sensor includes an insertable member, the insertable member extending through the base and insertable into the anatomy upon deployment of the physiological sensor, wherein the gravity resistance system extends at least partially around the insertable member.
17. A method of making a system, the method comprising: providing a physiological sensor having a base; coupling the physiological sensor to a movable support of an inserter, the movable support being movable to deploy the physiological sensor; coupling an adhesive patch to the base of the physiological sensor, the adhesive patch having a first surface and a second surface opposite the first surface, the second surface configured to couple the physiological sensor to an anatomy upon deployment; and coupling a first tack adhesive of a gravity resistance system to the support of the inserter; coupling a second tack adhesive of the gravity resistance system to the first surface of the adhesive patch, the second tack adhesive being less tacky than the first tack adhesive, and the gravity resistance system being spaced apart from the base; and maintaining, with the gravity resistance system, at least a peripheral portion of the adhesive patch adhered to the support prior to deployment of the physiological sensor and is removable from the adhesive patch upon deployment of the physiological sensor.
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